Code No: 134BU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, November/December - 2020
OPERATING SYSTEMS
(Common to CSE, IT)

Time: 2 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1. What are system calls? Explain about the various types of systems calls used in OS. [15]

2.a) Explain about the services of operating system.
b) Explain the O.S Structure. [8+7]

3. Consider Five processes P1 to P5 arrived at same time. They have estimated running time 10,2,6,8 and 4 seconds respectively. Their priorities are 3,2,5,4 and 1 respectively with 5 being highest priority. Find the average turn around time and average waiting time for Round Robin(q=3) and priority scheduling algorithm. [15]

4. What is Semaphore? Give the implementation of producer-consumer problem using Semaphore. [15]

5.a) Explain the concept of segmentation for memory management.
b) Explain the steps involved for handling the page fault. [8+7]

6.a) What is fragmentation? Explain the difference between internal and external fragmentation.
b) Explain about the structure of Page Table. [8+7]

7. Consider the following disk queen with requests for I/O to blocks on cylinders 98,183,37,122,14,124,65,67 in that order, using FCFS, SCAN algorithms of the disk head is initially at cylinder 53, find the total head movement in cylinders. Also provide the necessary diagram to show the head movement for the above queue. [15]

8. What do you mean by deadlock avoidance? Explain the use of Banker's algorithm for deadlock avoidance with illustration. [15]

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